Experiment 5: Real-Time Streaming Data Processing using Databricks & Azure Event Hub

Course: Cloud Analytics (25MCAR0141) | Course Outcome: CO3

Aim

To process real-time streaming data using Azure Databricks and Azure Event Hub by ingesting a live data stream, processing it using Structured Streaming in PySpark, and displaying the output in a Databricks notebook.

Objectives

- Create and configure an Azure Event Hub namespace and entity with SAS policies.
- Implement a Python producer script that simulates real-time student activity events.
- Read the stream in Azure Databricks using the Event Hub Spark connector.
- Parse JSON event payloads and apply schema enforcement.
- Perform tumbling window aggregations (event counts, average scores per student).
- Write streaming output to ADLS Gen2 in Parquet format with checkpointing.

Architecture & Component Roles

Resource Group: 25MCAR0141

Logical container for all Azure experiment resources

Event Hub Namespace: calab-eventhub-ns-141

Managed event ingestion service (Basic SKU, centralindia)

Event Hub: calab-eventhub-141

Event stream entity (2 partitions, 1-hour retention)

SAS Policy: DatabricksPolicy

Shared Access Policy with Send + Listen permissions

Databricks Workspace: calab-databricks-141

Spark analytics platform running Structured Streaming notebooks

Storage Account: calablake141

ADLS Gen2 - output sink for streaming Parquet files

Architecture Diagram

```
Local Python producer (producer.py)
        |  50 student activity events (JSON)
        v
Azure Event Hub Namespace (calab-eventhub-ns-141)
        |  Event Hub entity, 2 partitions
        v
Azure Databricks (calab-databricks-141)
        |  Structured Streaming + Event Hub Spark connector
        |  Parse JSON -> schema enforcement
        |  Tumbling window (5 min) + watermark (2 min)
        v
ADLS Gen2 (calablake141) -> Parquet output + checkpoint
        v
```

```
output/streaming_events
```

Step-by-Step Execution Summary

- Step 1: Create Event Hub Namespace: Azure Portal -> Event Hubs -> Create Namespace calab-eventhub-ns-141 in RG 25MCAR0141.
- Step 2: Create Event Hub Entity: Inside namespace -> + Event Hub -> Name: calab-eventhub-141, Partitions: 2.
- Step 3: Create SAS Policy: Event Hub -> Shared Access Policies -> Add DatabricksPolicy with Send + Listen.
- Step 4: Copy Connection String: Copy the Primary Connection String from DatabricksPolicy (paste into the notebook placeholder only; do not commit a real key).
- Step 5: Open Databricks Studio: Navigate to calab-databricks-141 -> Launch Workspace.
- Step 6: Create Cluster: Compute -> Create Cluster -> Single Node, Runtime 14.3 LTS, Standard_DS3_v2.
- Step 7: Install Maven Library: Cluster -> Libraries -> Install New -> Maven -> com.microsoft.azure:azure-eventhubs-spark_2.12:2.3.22.
- Step 8: Import Streaming Notebook: Upload Exp05_Databricks_Streaming.py -> paste connection string.
- Step 9: Run Producer Script: Locally run python producer.py to send 50 student activity events.
- Step 10: Start Streaming Notebook: Run all cells - observe real-time events and windowed aggregations.
- Step 11: Verify Output: Check calablake141/output/streaming_events for Parquet files.
- Step 12: Stop Stream & Terminate Cluster: Stop streaming query, terminate cluster to save cost.

Code Highlights

Producer - Send Events to Event Hub (full script in producer.py):

```
from azure.eventhub import EventHubProducerClient, EventData
event = {"student_name": "Abhi", "event_type": "quiz_submit", "score": 85, ...}
producer.send_batch(event_data_batch)
```

Consumer - Structured Streaming from Event Hub (full notebook in Exp05_Databricks_Streaming.py):

```
raw_stream = spark.readStream.format("eventhubs").options(**ehConf).load()
parsed = raw_stream.select(from_json(col("body").cast("string"), schema).alias("event"))
```

Tumbling Window Aggregation:

```
windowed = parsed.withWatermark("enqueued_time", "2 minutes") \
    .groupBy(window("enqueued_time", "5 minutes"), "student_name") \
    .agg(count("*").alias("event_count"), avg("score").alias("avg_score"))
```

Screenshot Checklist

- 1. Azure Portal - Event Hub Namespace overview (Step_01_*.png - see README for exact filenames)
- 2. Event Hub entity calab-eventhub-141 details (Step_02_*.png - see README for exact filenames)
- 3. SAS Policy DatabricksPolicy with keys (Step_03_*.png - see README for exact filenames)
- 4. Databricks Cluster with Maven library installed (Step_04_*.png - see README for exact filenames)
- 5. Producer script running - terminal output showing events sent (Step_05_*.png - see README for exact filenames)
- 6. Raw streaming events displayed in notebook (Step_06_*.png - see README for exact filenames)
- 7. Parsed JSON stream with student names and event types (Step_07_*.png - see README for exact

filenames)
- 8. Tumbling window aggregation - events per student (Step_08_*.png - see README for exact filenames)
- 9. Event type distribution streaming chart (Step_09_*.png - see README for exact filenames)
- 10. Streaming query status (active) (Step_10_*.png - see README for exact filenames)
- 11. Output Parquet files in ADLS Gen2 (Step_11_*.png - see README for exact filenames)

Result

A live student-activity stream was ingested through Azure Event Hub, parsed and schema-enforced in Databricks Structured Streaming, aggregated into 5-minute tumbling windows per student (event count and average score), and written to ADLS Gen2 as Parquet files with checkpointing.

Key Learning

This experiment demonstrates a complete real-time streaming pipeline on Azure: an event producer feeding Event Hub, a Spark Structured Streaming consumer that parses, validates and window-aggregates the stream, and a durable Parquet sink with checkpointing for fault-tolerant, resumable processing.